



## Full Length Article

## Sensitivity analysis of the inputs for bioactivity-exposure ratio calculations in a NAM-based systemic safety toolbox

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## ABSTRACT

To support regulatory decision-making without animal testing, Next-Generation Risk Assessment (NGRA) frameworks leverage New Approach Methodologies (NAM). A widely used strategy in such frameworks is to calculate Bioactivity-Exposure Ratios (BERs), which compare NAM assay-derived points of departure (PODs) and estimated human internal exposures. However, key methodological choices in NGRA, such as toxicokinetic modeling software, POD sources, and adjustments for *in vivo* and *in vitro* free concentrations, may introduce uncertainty. To determine the robustness of our previously reported NGRA workflow, we conducted a comprehensive sensitivity analysis of these variables as applied to 35 chemicals covering a range of end uses including consumer, pesticide and pharmaceutical. Physiologically-based kinetic (PBK) modeling using both commercial (GastroPlus) and open-source (*htk*) software was evaluated across three levels of parameterization. The *htk* model produced individual and population  $C_{max}$  estimates that are comparable to GastroPlus when the chemical-specific input parameters are aligned. BERs were calculated using  $C_{max}$  estimates, previously reported *in vitro* PODs from human-relevant cell-based assays, ToxCast, and fixed values based on internal threshold of toxicological concern (iTTCs), with and without adjustment for unbound fractions in plasma and media. We found that BERs from human-relevant cell-based PODs and nominal concentrations best matched risk classifications, with improved predictive performance over iTTC or ToxCast-derived PODs. Consideration of population variability in PBK had limited impact on risk prediction. Adjustments for free concentrations did not enhance classification or *in vitro/in vivo* concordance. These findings support the use of human-relevant *in vitro* PODs and nominal dose metrics in NGRA, and demonstrate that accessible, well-parameterized PBK tools can enhance efficiency and reproducibility.

## 1. Introduction

Some regulatory actions limit or ban animal testing for certain types of chemicals; therefore, new approach methodologies (NAMs), including human-relevant *in vitro* assays, *in chemico*, and *in silico* approaches, are the only option for safety assessments of such chemicals (Escher et al., 2022). Accordingly, next-generation risk assessment (NGRA) has been proposed as a framework that integrates various types of NAMs-based data to inform safety decision-making without animal testing (Escher et al., 2022; Schmeisser et al., 2023). A number of roadmaps for integrating such data into risk evaluations have been proposed by the European Food Safety Authority (Escher et al., 2022), the U.S.

Environmental Protection Agency (EPA) (Thomas et al., 2019), and the International Cooperation on Cosmetics Regulation (ICCR) (Dent et al., 2018). Among these, the ICCR's roadmap for NGRA emphasizes that the risk assessment process should be human-relevant, exposure-led, hypothesis-driven, and designed to prevent harm. A recent review of the publications on NGRA by Vaillancourt et al. (2025) observed that most case studies that followed ICCR's principles focused on ensuring that chemicals do not elicit meaningful biological activity in NAMs-based studies (e.g., "protection"), rather than predicting a specific toxicological hazard.

Traditionally, quantitative risk assessment has relied on animal data to derive points of departure (PODs) that are compared with predicted

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human exposure levels to calculate the margin of exposure (MOE), which informs risk characterization (Barlow, 2016). NGRA approaches adopt the same conceptual framework. For example, Baltazar et al. (2020) aimed to predict risk with a protective approach. They reasoned that if human exposure to a chemical is below the concentration that elicits any biological effect in non-animal experimental models with sufficient biological coverage, adverse effects will be unlikely. They used a case study of coumarin in cosmetics in which exposure scenarios were defined based on consumer use, physiologically-based kinetic (PBK) models were used to estimate internal exposure (e.g., plasma) concentrations, and PODs were derived from dose-response data in human-relevant *in vitro* assays. Then, the *in vitro*-derived PODs and internal exposure estimates were used to calculate the bioactivity exposure ratio (BER), which is conceptually similar to margin of exposure (Paul Friedman et al., 2020). Subsequently, Middleton et al. (2022) proposed a generalizable NGRA workflow based on this approach with an associated toolbox of *in vitro-in silico* methods. Using 10 chemicals with established human exposure scenarios, BER values were calculated to derive NAMs-based chemical-/exposure-specific risk levels that were compared to historical safety decisions based on known human uses or conventional safety data. It was found that the proposed workflow and toolbox could be used to support risk-based decision making without studies in animals. More recently, Cable et al. (2025) applied this workflow and experimental methods to assess the safety of an additional 38 chemicals across 70 human exposure scenarios. This study further increased scientific confidence in the utility of the NGRA approach.

Encouraging findings notwithstanding, the NAM-based toolbox and workflow proposed by Middleton et al. (2022) integrates a number of experimental and computational methods that require making certain assumptions. These include choices of the PBK modeling software, selection of assays from which to derive *in vitro* PODs, and the selection of dose metrics used to relate *in vitro* and *in vivo* concentrations (e.g., using nominal or total versus unbound concentrations in plasma or test media while characterizing risk). For example, widely available PBK software packages may differ in model structure and parameterization including how they account for physiological variability (e.g., interindividual differences in metabolism, organ blood flow, plasma protein binding, and tissue partition coefficients), assumptions of linear kinetics, and formulation-related factors such as solubility and dissolution. This results in variability in predicted outputs such as concentrations in plasma or tissues (Ahmad et al., 2020) (Prieto Garcia et al., 2022; Zhang et al., 2025). Choices of *in vitro* models and how PODs are calculated can also influence BER (Paul Friedman et al., 2025). Studies by Middleton et al. (2022) and Cable et al. (2025) relied on several cell types for high-throughput transcriptomics PODs, as well as including phenotypic measurements and cell-free bioactivity data on key receptors, enzymes, transporters, and ion channels (Bowes et al., 2012). An alternative data source for thousands of chemicals is the U.S. EPA's Toxicity Forecaster (ToxCast) database, which includes high-throughput screening data (Williams et al., 2017) that cover a wide range of assays in a diverse set of models including human and animal cell lines, yeast, and cell-free systems. Another alternative for inferring *in vitro* PODs without new testing is to use a fixed threshold value based on the internal Threshold of Toxicologic Concern (iTTC) approach (Arnot et al., 2022; Blackburn et al., 2020; Ellison et al., 2020). Finally, the selection of appropriate dose metrics in NGRA and the application of quantitative *in vitro-to-in vivo* extrapolation (QIVIVE) to translate *in vitro* assay data into *in vivo* toxicity estimates are also subject to expert-defined assumptions (Henneberger et al., 2021; Nicol et al., 2024). For example, it has been suggested that the free concentration in the media of *in vitro* assays is a more appropriate metric for comparison with the freely dissolved concentration in plasma from *in vivo* studies (Fisher et al., 2019; Heringa et al., 2004). However, estimating *in vitro* and *in vivo* unbound (free) chemical concentrations in assay media and human plasma, respectively, requires additional resources or is subject to model choices that may introduce additional uncertainty (Lin et al., 2025; Proença et al.,

2021).

Because it has not been determined how certain assumptions, such as those detailed above, can affect the outcome of NGRA, additional analyses are needed to understand the robustness of the overall approach. Accordingly, this study conducted sensitivity analyses of the critical choices in the overall NGRA workflow as applied to chemicals and their consumer exposure scenarios previously published by Middleton et al. (2022) and Cable et al. (2025). Specifically, we evaluate whether different PBK modeling tools, sources of *in vitro* PODs, and the use of total or free concentrations in BER calculations impact the workflow-derived risk levels. Overall, this study aimed to enhance scientific confidence in the robustness of the NGRA workflow and to provide practical recommendations for its future use in decision-making contexts concerning chemical risk assessment.

## 2. Methods

### 2.1. Overview of the NGRA workflow

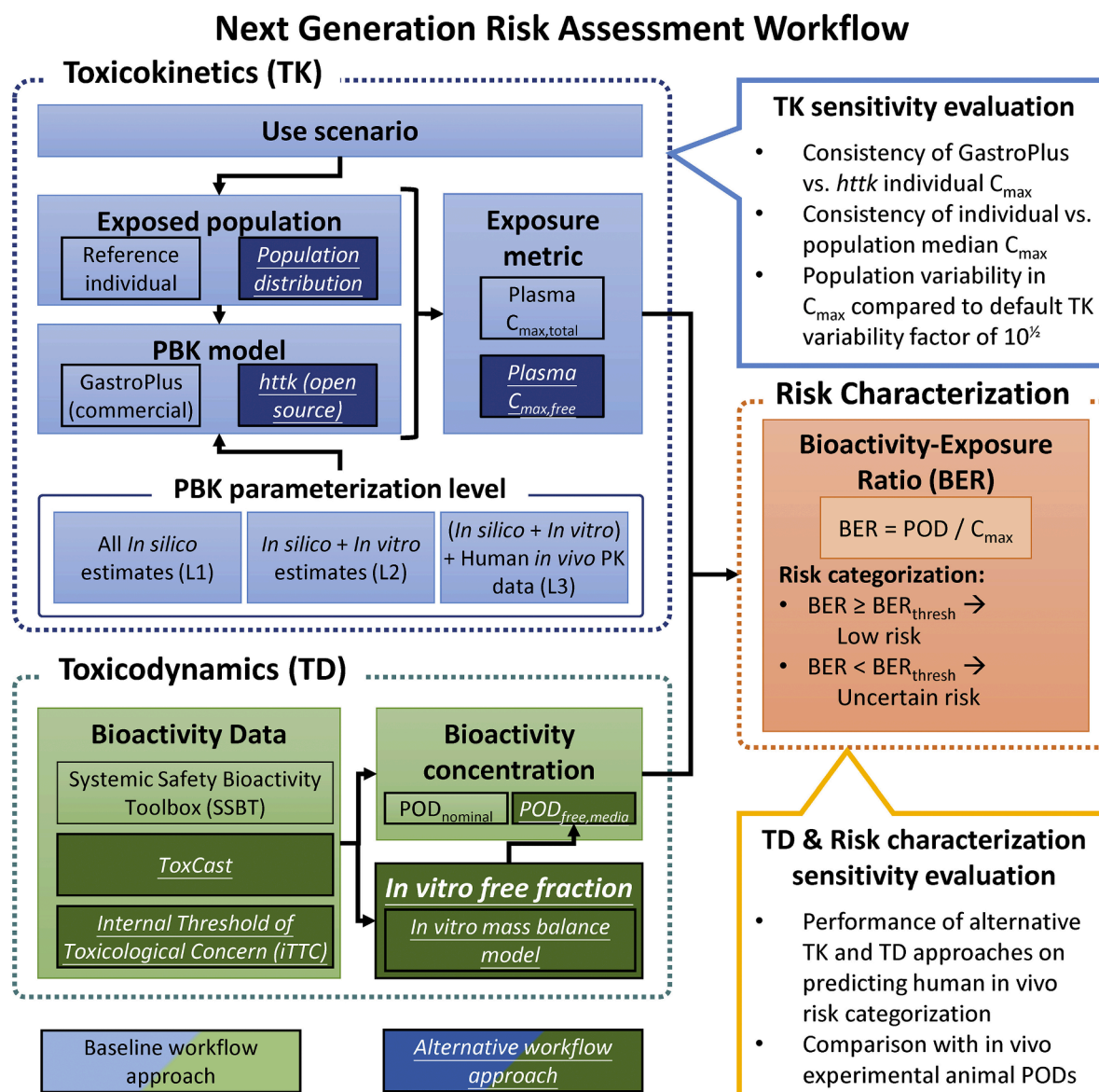
With a focus on developing NAMs to reduce animal testing, NGRA provides a framework for incorporating non-animal data into safety assessments to support risk-related decision-making. Middleton et al. (2022) outlined a systematic NGRA workflow comprising three main modules:

- Exposure assessment (Toxicokinetics [TK]): Estimating internal exposure to a chemical based on defined use-case scenarios.
- Toxicity assessment (Toxicodynamics [TD]): Deriving points of departure (PODs) using *in vitro* assays.
- Risk characterization: Integrating results from the TK and TD Modules to calculate the bioactivity-exposure ratio (BER) and determining potential risk levels.

This study follows the same workflow, as shown in Fig. 1.

The approach for each module is detailed in Middleton et al. (2022). Briefly, in the *exposure assessment* module, at least one consumer exposure scenario was defined for each target chemical. These scenarios served as input to PBK modeling to estimate the maximum internal concentration ( $C_{\max}$ ) of each chemical in plasma. Risk classifications (high or low) for each exposure scenario were assigned based on available toxicological data, following the rationale described by Middleton et al. (2022) and Cable et al. (2025). Parameterization of PBK models is a critical and complex step, as parameter values may come from *in silico* predictions, *in vitro* measurements, or *in vivo* data (when available, for calibration). Accounting for this uncertainty, PBK models were constructed at three different parameterization levels based on combinations of data sources (i.e., from *in silico*, *in vitro*, and *in vivo*), as proposed by Middleton et al. (2022) and Cable et al. (2025). Detailed descriptions of the three levels are provided in Section 2.2.1. In this way, the  $C_{\max}$  predictions depended on both the consumer exposure scenario and the parameterization level of the PBK model.

The *toxicity assessment* module, as described by Middleton et al. (2022) and Cable et al. (2025), involves generating dose-response *in vitro* bioactivity data to derive points of departure (PODs). The *in vitro* platform used by Middleton et al. (2022) and Cable et al. (2025), denoted here the Systemic Safety Bioactivity Toolbox (SSBT), includes high-throughput transcriptomics (HTTr), a cell stress panel (CSP), and *in vitro* pharmacological profiling (IPP) data. Different dose-response modeling approaches were applied to estimate PODs depending on the assay: for HTTr, Bayesian dose response modelling using a non-parametric model along with BMDexpress (Phillips et al., 2019) were used; for IPP, Bayesian dose response modelling using Hill functions was applied (Johnstone et al., 2016). In the *risk characterization* module, BERs were calculated using the minimal PODs across selected platforms in combination with the  $C_{\max}$  values estimated under each consumer exposure scenario and PBK model parameter level. For



**Fig. 1. Overall study workflow.** Illustration of sensitivity analysis within each of the three Next Generation Risk Assessment modules: toxicodynamics (TD), toxicokinetics (TK), and risk characterization. In each module, the baseline approaches are shown along with the alternative approaches in darker boxes (*italic and underlined*). Callout boxes describe the sensitivity evaluations done for each module. **Abbreviations:** TTC, threshold of toxicological concern; POD, point of departure;  $C_{max}$ , maximum plasma concentration; BER, bioactivity-exposure ratio.

BER-based decision-making, parameterization level-specific thresholds were applied to classify the risk level of each exposure scenario. These BER-based classifications were then compared to the original risk assignments from the exposure assessment module to evaluate the binary classification performance of the NGRA workflow.

Next, we evaluated how sensitive the risk classification from NGRA may be in response to methodological choices made in each module:

- **Exposure assessment:** Choice of PBK modeling software (commercial vs. open-source) and whether to use point estimates or population-level  $C_{max}$ .
- **Toxicity assessment:** Use of PODs derived from the NGRA cell-based assays, the ToxCast battery, or a fixed value based on the internal Threshold of Toxicological Concern (iTTC), as proposed by Blackburn et al. (2020) or Arnot et al. (2022).
- **Risk characterization:** Whether BER calculations accounted for *in vivo* and *in vitro* total or free concentrations based on the unbound fraction of the chemical in human plasma or test media.

Using these combinations, BERs were calculated and compared to the classification based on *in vivo* historical use or conventional safety assessments to evaluate which combination achieved the best risk classification performance. The comparison was based on a binary classification approach (using a BER threshold of 1) and an assessment of the correlation between BER values and literature-informed risk categories. We hypothesize that this sensitivity analysis provides valuable insights and practical guidance for future users of this NGRA workflow.

## 2.2. Exposure assessment: $C_{max}$ and toxicokinetic variability factor (TKVF) estimation

### 2.2.1. $C_{max}$ reproducibility using commercial and open-source tools

In the studies by Middleton et al. (2022) and Cable et al. (2025), maximum plasma concentrations ( $C_{max}$ ) following various exposure scenarios were predicted using PBK models implemented in commercial software (i.e., GastroPlus, SimulationsPlus, Lancaster, CA, ver. 9.8.0 and

9.9.0). The parameterization of these PBK models was categorized into three levels based on the complexity and data sources:

- i. Level 1 (L1) – *In silico* data only: All chemical-specific parameters were predicted solely through *in silico* methods.
- ii. Level 2 (L2) – Combination of *in silico* and *in vitro* data: Key chemical-specific parameters i.e., logP, pKa, solubility, hepatic intrinsic clearance ( $CL_{int}$ ), plasma unbound fraction ( $f_{ub,plasma}$ ), blood-to-plasma ratio, and intestinal absorption were primarily sourced from *in vitro* assays, when available. Remaining parameters were collected through *in silico* predictions (same as L1).
- iii. Level 3 (L3) – Combination of *in silico*, *in vitro* and *in vivo* data: Building on the L2 parameter set, *in vivo* clinical data were incorporated, if available, to calibrate the model and refine critical parameters. Further details on parameter selection and model calibration can be found in Middleton et al. (2022) and Cable et al. (2025).

Due to the time-intensive nature of conducting population-level PBK simulations in commercial software—where each individual chemical often requires a separate manual execution—open-source alternatives such as the *httk* R package (developed by the US EPA, ver 4.3.1) support automated batch simulation across chemicals, significantly improving efficiency (Pearce et al., 2017).

However, *httk* uses a different set of default parameter than those used in GastroPlus by Middleton et al. (2022) and Cable et al. (2025). Specifically, the *httk* R package includes an internal database of chemical-specific parameters ( $CL_{int}$ ,  $f_{ub,plasma}$ , blood-to-plasma ratio, and intestinal absorption) and calculates partition coefficients using the method of Schmitt (Schmitt, 2008), based on logP and pKa, which is also different from the parameters that Middleton et al. (2022) and Cable et al. (2025) collected to use in GastroPlus. These differences in parameter sources and estimation methods may lead to differences in predicted  $C_{max}$  values between *httk* and GastroPlus.

To evaluate whether *httk* can approximate  $C_{max}$  estimates from previously published GastroPlus models (parameterized using the L1–L3 strategies) (Middleton et al., 2022, Cable et al., 2025), we systematically tested four different parameter replacement strategies within *httk*, and compared the resulting  $C_{max}$  values against those from GastroPlus (both using perfusion-limited models):

- (A) Using the default *httk* parameter set (i.e., all available chemical-specific parameters and physiological parameters from the *httk* database).
- (B) Starting from the *httk* default, we replaced key chemical-specific parameters (logP, pKa,  $f_{ub,plasma}$ , blood-to-plasma ratio,  $CL_{int}$ , and gut absorption rate) with values matching those used in each of the GastroPlus L1–L3 parameter sets (note that gut absorption rate was based on the lumped one-compartment value from GastroPlus).
- (C) Same as (B), but without adjusting the *httk* gut absorption rate.
- (D) Same as (B), but we also replaced physiological parameters and tissue partition coefficients ( $K_p$ ) with those used in each of the GastroPlus L1–L3 parameter sets.

Given that the *httk* package offers greater flexibility for modeling oral exposure routes compared to dermal, intravenous or inhalation routes and considering the parameter availability from the *httk* database, this reproducibility assessment included only the 35 chemicals for which oral exposure scenarios were defined. Detailed information on the parameter sets and exposure scenarios used in both GastroPlus and *httk* are provided in the Tables S1–S3. The version of *httk* used in this study was 2.3.1.

To evaluate which parameter replacement strategy in *httk* showed the greatest consistency with GastroPlus predictions (as reported in Middleton et al. (2022) and Cable et al. (2025)), we calculated the mean

absolute error (MAE) and mean error (ME) using the following equations:

$$MAE = \frac{\sum |\log_{10}(C_{max}^{httk}) - \log_{10}(C_{max}^{GastroPlus})|}{N} \quad (1)$$

$$ME = \frac{\sum (\log_{10}(C_{max}^{httk}) - \log_{10}(C_{max}^{GastroPlus}))}{N} \quad (2)$$

where  $C_{max}$  values are standardized by exposure dose (mg/L per mg/kg-d). In addition, Spearman's rank correlation coefficient ( $\rho$ ) and Pearson's correlation coefficient ( $r$ ) were also calculated to assess the association between the  $C_{max}$  predictions from the two modeling approaches.

Although comparing model-predicted  $C_{max}$  values to measured clinical data could provide valuable insights, such comparisons were not feasible in this study due to the limited availability of human clinical  $C_{max}$  values. As reported in Middleton et al. (2022), only five chemicals across six oral exposure scenarios had reported  $C_{max}$  values. Despite this limitation, Middleton et al. (2022) still included such comparisons in their analysis to illustrate model performance across using different parameterization levels.

## 2.2.2. Population-level simulations and derivation of TKVF

Population variability in  $C_{max}$  estimates is a key consideration when conducting exposure assessments. To simulate  $C_{max}$  with population-level variability, we used the *httkpop* function—a virtual population generator within the *httk* package (population simulations were not performed using GastroPlus). Specifically, we created a virtual population of 1,000 individuals aged between 16 and 79 years, with a male-to-female ratio of 504:496 (Ring et al., 2017). The detailed physiological parameters for this virtual population are listed in Table S4. The parameterization applied in this population simulation was based on the optimized parameter set that demonstrated the highest consistency in  $C_{max}$  predictions between GastroPlus and *httk*.

Chemical-specific parameters such as  $f_{ub,plasma}$  and  $CL_{int}$  can vary across individuals. These parameters were therefore considered in the Monte Carlo simulation to reflect population variability (Pearce et al., 2017). We modeled variability in  $f_{ub,plasma}$  using a truncated normal distribution with the chemical-specific input value as the mean, a CV of 0.1, and a lower truncation bound of  $10^{-4}$  (the default minimum value in *httk*). The chosen CV is based on the results from Fischer et al. (2024), who analyzed the impact of population variability in serum components such as albumin for plasma protein binding of PFAS and found that across a range of PFAS containing 4–13 carbons, the coefficient of variation (CV) for  $f_{ub,plasma}$  was always less than 10%.  $CL_{int}$  was assumed to be normal distributed, with the chemical-specific input value as the mean and CV values of 0.1, 0.3, and 0.5 evaluated separately, and a lower bound truncated at zero. This population-level  $C_{max}$  estimation included the 35 chemicals for which oral exposure scenarios were defined.

It is typical in risk assessment and hazard characterization to use a default 10-fold “uncertainty” or “assessment” factor to address differences in sensitivity across the human population, further divided into separate default toxicokinetic and toxicodynamic factors of 3.16-fold ( $10^{1/2}$ ) (WHO/IPCS, 2005). Here, a toxicokinetic variability factor (TKVF) was used to quantify population variability in internal dose metrics as a chemical-specific replacement for the default factor. After generating the distribution of internal  $C_{max}$  values, TKVF was calculated as the ratio of the  $C_{max}$  in a “sensitive” individual (e.g., at the 95th or 99th percentile) to the median  $C_{max}$  of the population (a general individual). Depending on the percentile chosen, TKVF<sub>95</sub> and TKVF<sub>99</sub> were derived and compared to the default value of 3.16.

## 2.3. In vitro data for determining chemical bioactivity and PODs

To evaluate whether the source of chemical bioactivity data

influences risk characterization, both cell assay-based (from the assay compendia as detailed below) and a fixed constant *in vitro* PODs (Arnot et al., 2022; Blackburn et al., 2020) were compared. The SSBT-based PODs were obtained from publications by Middleton et al. (2022) and Cable et al. (2025). Both studies used the identical set of *in vitro* assays (but different chemicals) to derive PODs; these included a cell stress panel (CSP) using HepG2 cells, high-throughput transcriptomics (HTTr) using HepG2, HepaRG, and MCF7 cells, and an *in vitro* pharmacological profiling platform (IPP) that evaluated chemical effects on 63 targets known to be associated with safety concerns, using binding, enzymatic, coactivator recruitment, and luciferase assays (Cable et al., 2025; Middleton et al., 2022). For both the CSP and HTTr assays, the BIFROST method (Bayesian Inference for Region of Signal Threshold) was applied to estimate a “global POD,” which is the minimum effect concentration across all genes (transcriptomics) or biomarkers (cell stress). From HTTr, an additional POD was calculated using BMDExpress2 which corresponded to the minimum pathway BMDL (1 for each cell line). However, for this subset of chemicals, these PODs were never the minimum and therefore not included in the present analysis. For the IPP data, for those compounds and targets that had an activity level of 50% compared to the matched assay positive control at 10 or 100  $\mu\text{M}$  in an initial screening, Bayesian concentration–response modeling was applied to derive  $\text{EC}_{50}$  (half-maximal effective concentration) or  $\text{IC}_{50}$  (half-maximal inhibitory concentration) values. The lowest  $\text{AC}_{50}$  (half-maximal activity concentration) from these assays was selected as the POD. Overall, this set of cell assay-based PODs included: (1) the lowest  $\text{AC}_{50}$  from the pharmacological profiling platform, (2) global PODs from transcriptomics data for each of the three cell lines, and (3) the global POD from the HepG2 cell stress panel (Table S5). Detailed experimental information and methods for POD derivation are described in Middleton et al. (2022) and Cable et al. (2025).

The second set of cell assay-based PODs was from the Toxicity Forecasting (ToxCast) database (Williams et al., 2017). ToxCast includes high-throughput *in vitro* screening assays for nearly 10,000 chemicals, assessing various biological processes such as mitochondrial function, development, neurotoxicity, and cell cycle progression, as well as molecular targets including nuclear receptors, enzymes, and gene expression pathways. For the 35 target chemicals evaluated in this study,  $\text{AC}_{50}$  values that passed the hit call (labeled “active”) were available for 29, and collected from the CompTox Dashboard (<https://comptox.epa.gov/dashboard/>, accessed 1-Jun-2025 [version 2.5.3, invitroDBv4.1]; Table S6). The 5<sup>th</sup> percentile of all active  $\text{AC}_{50}$  values for each chemical was used as the ToxCast-derived POD for this dataset.

Finally, fixed constant *in vitro* PODs were also included to isolate the impact of toxicokinetics on risk characterization. This approach allows for evaluation of how toxicokinetics alone impacts risk classification performance. A constant *in vitro* POD of 1  $\mu\text{M}$ , based on the internal Threshold of Toxicological Concern (iTTC) proposed by Blackburn et al. (2020) for non-pharmacologically active compounds, was applied. Additionally, we included the iTTC proposed by Arnot et al. (2022) that is not restricted to non-pharmacologically active compound, using the 0.01  $\mu\text{M}$  “internal NOAEL” value prior to applying a 100-fold safety factor.

#### 2.4. *In vivo/in vitro* free concentrations: $f_{\text{ub}}$ measurements and predictions

As part of this study’s approach, one objective was to assess whether incorporating *in vivo* and *in vitro* adjustments for free vs. nominal or total concentrations affects the performance of risk classification. *In vivo* free fraction is represented by the  $f_{\text{ub,plasma}}$ , a parameter used in the PBK model described above. *In vitro* free fraction was characterized by the fraction unbound in media ( $f_{\text{ub,media}}$ ), which can be predicted using the *in vitro* mass balance model (Armitage et al., 2021). Based on the study conducted by Lin et al. (2025), the prediction performance comparisons among several developed models showed that the Armitage et al. (2021)

model is a sensible first choice due to its relatively high predictive accuracy and low bias. The parameters required for the Armitage model were grouped into three categories: chemical- (molecular weight [MW], melting point [MP], octanol-water partition coefficient [ $K_{\text{OW}}$ ], air-water partition coefficient [ $K_{\text{AW}}$ ], solubility, pKa, and ionizable organic chemicals [IOC] type), cell- (protein, lipid and water content), media- (% of fetal bovine serum [FBS]) and labware- (well-plate material) related parameters. Most chemical-specific parameters were sourced from the Exposure and Safety Estimation (EAS-E) Suite (ARC Arnot Research and Consulting, <https://arnotresearch.com/eas-e-suite/>). The pKa and IOC type were predicted using the web-based tool developed by Pan et al. (2021) (Table S7). The chemical-specific parameters are listed in Table S8. Cell, media, and labware parameters were based on the *in vitro* assay protocols from the original studies (Table S9). Notably, because the cell types and percentage of fetal bovine serum (FBS) used in the assays measuring the lowest  $\text{AC}_{50}$  values from *in vitro* pharmacological profiling (Cable et al., 2025; Middleton et al., 2022) and the 5<sup>th</sup> percentile  $\text{AC}_{50}$  from ToxCast were not available, default cell parameters from the Armitage et al. (2021) model were used. Also, it was assumed that 10% FBS was used and that the assays were conducted in standard 384-well plates. The predicted  $f_{\text{ub,media}}$  values for different *in vitro* assay endpoints, based on the model developed by Armitage et al. (2021), are compiled in Table S10.

In addition to the  $f_{\text{ub,plasma}}$  values from the PBK model and the predicted  $f_{\text{ub,media}}$  values from the Armitage model, experimentally measured values of  $f_{\text{ub}}$  in both plasma and media for a subset of chemicals were also obtained from Nicol et al. (2024). The percentage of chemical bound to protein was calculated using the following equation:

$$\% \text{Bound} = \frac{C_{\text{matrix}} - C_{\text{supernatant}}}{C_{\text{matrix}}} \times 100 \quad (3)$$

where  $C_{\text{matrix}}$  and  $C_{\text{supernatant}}$  refer to the LC-MS responses measured in matrix (i.e., human plasma, HepG2, MCF7, and HepRG media) and supernatant, respectively. The reported plasma protein binding ranged from 10% to 99%, with values outside this range reported as either <10% or >99% bound. The percentage fraction unbound in plasma is calculated by  $100\% - \% \text{Bound}$  (from equation 3). For the censored values, <10% was replaced with 5% (the midpoint between 0% and 10%), and >99% was replaced with 99.5% (the midpoint between 99% and 100%). The resulting  $f_{\text{ub}}$  in plasma and media collected from Nicol et al. (2024) are included in the Table S11.

#### 2.5. Risk characterization

BERs were calculated as the ratios of *in vitro* PODs to the estimated  $C_{\text{max}}$ . Depending on the choices of  $C_{\text{max}}$ , *in vitro* POD, and whether the  $f_{\text{ub}}$  in plasma and media were considered, BERs could be calculated using different combinations:

- $C_{\text{max}}$  from GastroPlus point estimates or 95th percentile estimates from *httk-pop* ( $\text{CL}_{\text{int}}$  CV = 0.3 and  $f_{\text{ub,plasma}}$  CV = 0.1; see section 2.2.2 for the rationale), for a given chemical-exposure scenario and each PBK parameterization level;
- *In vitro* PODs from cell assays (minimum from the SSBT; 5<sup>th</sup> percentile from ToxCast) or fixed values; and
- Calculations with or without accounting for  $f_{\text{ub}}$  in plasma and media—i.e., the POD and  $C_{\text{max}}$  can be either total concentrations or free concentrations, estimated using all predicted  $f_{\text{ub}}$  values, or using measured  $f_{\text{ub}}$  where available (with the rest predicted). These cases are labeled as “Total,” “Pred,” and “Meas. + Pred.”

It should be noted that in the combination using fixed *in vitro* POD values with consideration of  $f_{\text{ub}}$  in media, the calculation assumed that the PODs from SSBT, including the lowest  $\text{AC}_{50}$  from the pharmacological profiling platform, global PODs from transcriptomic data for

three cell lines, and the cell stress panel, were replaced with fixed values (1 or 0.01  $\mu\text{M}$ ), and then multiplied by the  $f_{ub}$  in the corresponding media.

To evaluate how well BERs correlate with risk classifications, we used the Point-Biserial correlation coefficient (Das Gupta, 1960) – a statistical measure that quantifies the relationship between a binary variable and a continuous variable – to assess the relationship between the BERs and reported risk classifications. Middleton et al. (2022) and Cable et al. (2025) defined each consumer exposure scenario as “high” or “low” risk for safety decision-making based on the literature. Due to the negative correlation between BER and risk level, we set high risk = 0 and low risk = 1 when computing the Point-Biserial correlation coefficient. This coefficient was used to examine the relationship between the minimal BERs and the reported risk classifications. A higher coefficient for a given BER calculation method indicates better performance in distinguishing between high- and low-risk classifications.

Binary classification evaluation was also performed using a BER threshold of 1. If the BER was greater than 1, the predicted risk level was classified as low; otherwise, as high. By comparing the predicted and reported risk classifications, classification performance metrics, namely accuracy, sensitivity, specificity, and Matthews correlation coefficient (MCC), were calculated using the equations described by Chicco and Jurman (2020). Except for MCC, which ranges from -1 to 1, all other metrics range from 0 to 1. A BER calculation method is considered to have better risk classification performance if it demonstrates high sensitivity, specificity, positive and negative predictive values, and MCC, along with low false positive and false discovery rates (Chicco et al., 2021).

## 2.6. Evaluation of *in vitro-in vivo* concordance of oral equivalent dose (OED) using different bioactivity data

In addition to accounting for *in vivo* and *in vitro* free concentrations in risk characterization, such adjustments have been proposed to improve the concordance between *in vitro* and *in vivo* OED estimates through quantitative *in vitro-to-in vivo* extrapolation (QIVIVE) (Armitage et al., 2021). For examining this hypothesis, we applied the human-equivalent doses (HEDs) collected from Cable et al. (2025) and Aurisano et al. (2023) (Table S12), along with *in vitro* POD datasets derived from cell assays and fixed constant values. The HEDs were further extrapolated from the animal PODs using allometric scaling and adjustment for exposure duration through uncertainty factors, where applicable. The animal PODs used in Cable et al. (2025) were derived from *in vivo* studies (units: mg/kg/day) and sourced from ToxRefDB (version 2.1, <https://www.epa.gov/comptox-tools/downloadable-computational-toxicology-data>), the supplementary material of Paul Friedman et al. (2020), and published regulatory sources. Additional *in vivo* animal PODs were obtained from the dataset published by Aurisano et al. (2023). Values for both animal and human-equivalent PODs, along with extrapolation methods and experimental context, are provided in Table S12.

The *in vivo* PODs were expressed in mg/kg/day, representing administered doses and thus interpretable as OEDs, allowing them to be directly compared to OEDs extrapolated from *in vitro* PODs. *In vitro* PODs were converted to OEDs using the QIVIVE approach, with the equation:

$$\text{OED (mg/kg/day)} = \text{POD}_{in\ vitro}(\mu\text{M}) \times \frac{1\text{ mg/kg/day}}{C_{ss, 1\text{ mg/kg/day}}(\mu\text{M})} \quad (4)$$

where  $C_{ss, 1\text{ mg/kg/day}}$  is the steady-state plasma concentration under daily oral dosing of 1 mg/kg/day, calculated using the *calc\_analytic\_css* function with the “3compartmentss” model and the settings adjusted.Fun-bound.plasma = FALSE and adjusted.Clint = FALSE in the *httk* R package (version 2.3.1). To account for  $f_{ub}$  in plasma and media, adjusted OEDs were calculated as:

$$\text{OED}_{adjusted} \text{ (mg/kg/day)} = \left( \text{POD}_{in\ vitro} \times f_{ub, media} \right) \times \frac{1\text{ mg/kg/day}}{C_{ss, 1\text{ mg/kg/day}} \times f_{ub, plasma}} \quad (5)$$

The values of  $f_{ub, plasma}$  were sourced from the PBK model, while those for  $f_{ub, media}$  were predicted using the Armitage model. Concordance between *in vitro*- and *in vivo*-derived OEDs was evaluated using Pearson’s  $r$ , Spearman’s  $\rho$ , MAE, and ME, as previously described.

## 2.7. Data and programming repository

The datasets used in this study and the R scripts implementing the analyses are available in the supplementary materials of this publication and the GitHub repository at [https://github.com/hsingchiehlin/Se-nana\\_NGRA](https://github.com/hsingchiehlin/Se-nana_NGRA).

## 3. Results

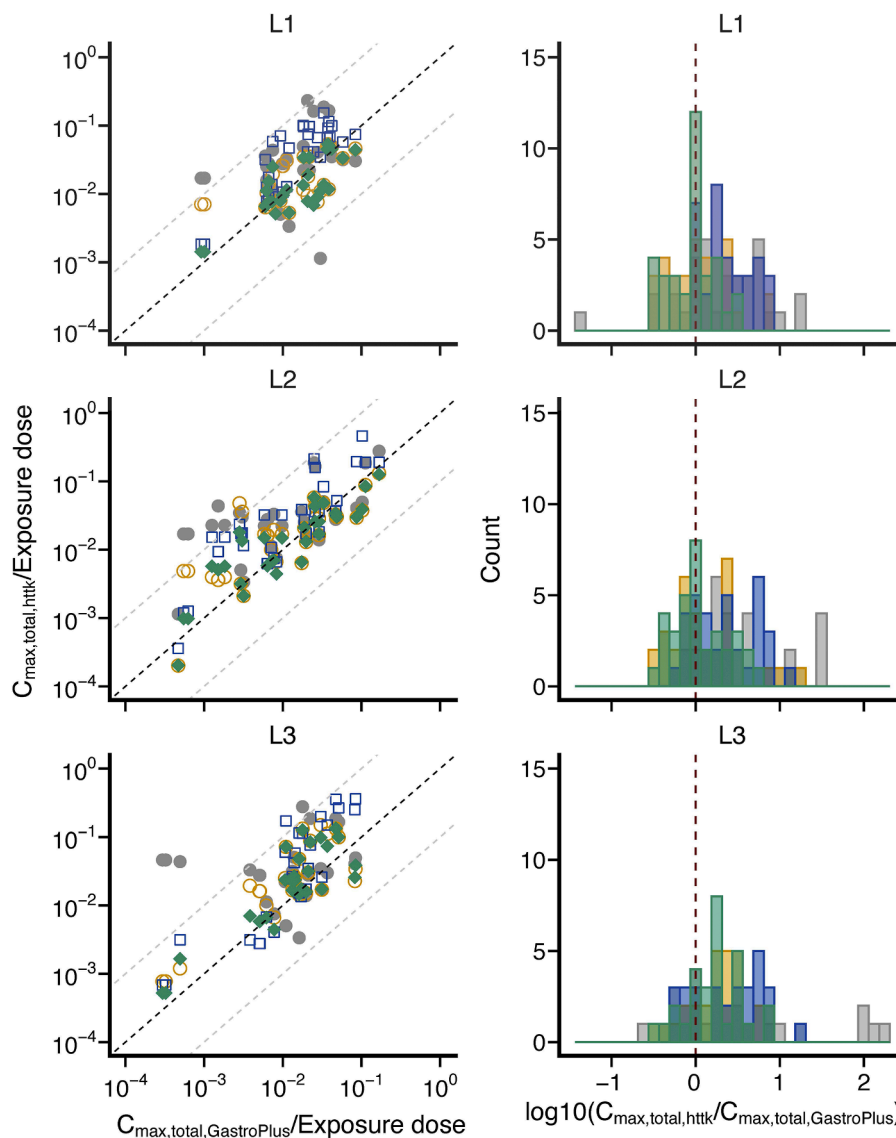
### 3.1. Reproducibility of $C_{max}$ estimates across PBK parameterizations

$C_{max}$  values reported in Middleton et al. (2022) and Cable et al. (2025) were derived using the commercial software GastroPlus. We tested if *httk* could produce  $C_{max}$  estimates comparable to those obtained with GastroPlus. The predicted  $C_{max}$  values from GastroPlus were extracted directly from published studies (Cable et al., 2025; Middleton et al., 2022), while the tested parameter sets were constructed by combining inputs from both GastroPlus and the *httk* default database. We applied four parameter sets in *httk*, and compared the resulting  $C_{max}$  estimates against previously reported GastroPlus values to assess consistency. Fig. 2 demonstrates the comparisons of external dose-standardized  $C_{max}$  values across tools and parameter combinations, with histograms showing the discrepancies. The detailed data used in Fig. 2 are compiled in Table S13. Table 1 summarizes performance metrics, including Spearman’s  $\rho$ , Pearson’s  $r$ , mean absolute error (MAE), and mean error (ME), for each comparison.

Among the three levels of PBK parameterization, all  $C_{max}$  estimates using three different parameter combinations modified from *httk* defaults remained within one order of magnitude of each other when compared to GastroPlus estimates (Fig. 2). In particular, when logP, pKa,  $f_{ub, plasma}$ , blood-to-plasma ratio,  $CL_{int}$ , and gut absorption rate are harmonized in *httk* with those previously defined in GastroPlus (parameterization approach (B)), high correlation coefficients ( $\rho$  and  $r$ ) are obtained and the lowest MAE, and ME values closest to zero (Table 1), indicating high consistency with GastroPlus. Excluding the gut absorption rate from the parameters that were harmonized with GastroPlus resulted in a reduced correlation, higher MAE, and greater ME deviation (Table 1). When the physiological parameters and partition coefficients from GastroPlus were included in *httk*, both  $\rho$  and  $r$  values remained the same, but MAE and ME increased, suggesting a decrease in  $C_{max}$  concordance despite extra complexity (Table 1). Overall, across all levels of PBK parameterization, the parameter set using *httk* defaults except for the five key chemical-specific parameters of logP,  $f_{ub, plasma}$ , blood-to-plasma ratio,  $CL_{int}$ , and gut absorption rate was selected for subsequent analyses as the best balance between complexity (i.e., how many parameters are included) and reproducibility of  $C_{max}$  estimates.

### 3.2. Population-level estimates of $C_{max}$ and TKVF derivation

Using the *httk* default parameter set with modifications to five key parameters (logP,  $f_{ub, plasma}$ , blood-to-plasma ratio,  $CL_{int}$ , and gut absorption rate), population-level  $C_{max}$  values were simulated. First, we compared the median  $C_{max}$  predictions from *httk-pop* with those from GastroPlus to ensure that population-level simulation did not change



**Fig. 2. Comparisons of maximum plasma concentrations ( $C_{\max}$ ) predicted using GastroPlus and *httk* R package.** The analyses were based on three levels of PBPK parameter integration (L1-L3, see Methods for details). In the scatter plots (left panels),  $C_{\max}$  values are standardized by exposure dose (mg/L per mg/kg-d). Gray solid circles (●) represent predictions from *httk* using default parameters from *httk* database. Green solid diamonds (◆) indicate predictions adjusting five parameters (logP, fraction unbound in plasma, blood-to-plasma ratio, hepatic intrinsic clearance, and the gut absorption rate) to those used in GastroPlus. Yellow open circles (○) represent predictions with adjusted parameters from GastroPlus but without modifying the gut absorption rate. Blue open squares (□) reflect predictions using *httk* with additional parameters from GastroPlus, including the five adjusted parameters, physiological parameters, and partition coefficients. The histograms (right panels) display the differences in standardized  $C_{\max}$  values between GastroPlus and *httk* predictions. Dashed lines indicate where the fold-difference is 1 (black, all panels) or 10-fold in either direction (grey, left panels only).

consistency. **Fig. S1** shows these comparisons across different PBK parameterization levels and  $CL_{\text{int}}$  CVs, along with the correlation coefficients ( $\rho$  and  $r$ ), MAE, and ME. Compared to the results in **Table 1**, **Fig. S1** demonstrates that, regardless of PBK parameterization level or  $CL_{\text{int}}$  variability, *httk-pop* predictions had slightly higher correlation coefficients, lower MAEs, and MEs were closer to zero compared to individual-level estimates. These results suggest that *httk-pop* was not only consistent with the GastroPlus  $C_{\max}$  values, but even slightly improved the concordance compared to individual-level estimates.

The median, 95<sup>th</sup> percentile, and 99<sup>th</sup> percentile of  $C_{\max}$  were then used to calculate the TKVF<sub>95</sub> and TKVF<sub>99</sub> values (**Fig. 3**; **Table S14**). In general, TKVF<sub>99</sub> values were higher than TKVF<sub>95</sub>, and many exceeded the default threshold of  $10^{0.5}$  ( $\approx 3.16$ ). Most TKVF<sub>95</sub> values across PBK parameterization levels and  $CL_{\text{int}}$  CVs were below this threshold, except for the 97.5<sup>th</sup> percentiles under  $CL_{\text{int}}$  CV = 0.5 for L2 and L3 PBK levels.

In contrast, the 97.5<sup>th</sup> (or 75<sup>th</sup>) percentile TKVF<sub>99</sub> values exceeded 3.16 in all cases except for L1 PBK parameterization with  $CL_{\text{int}}$  CV = 0.1 and 0.3. Additionally, TKVF values increased with higher levels of PBK parameterization.

### 3.3. Comparisons of NAM-based BERs

BERs were calculated using combinations of estimated  $C_{\max}$  (from GastroPlus or the 95<sup>th</sup> percentile from *httk-pop*), *in vitro* PODs (from cell assays or fixed values), and whether  $f_{\text{ub}}$  in plasma and media were considered. **Fig. 4** shows an example of BER calculations comparing total  $C_{\max}$  values from GastroPlus and *httk-pop*, under defined high- and low-risk exposure scenarios, and *in vitro* nominal PODs from SSBT. In this case, neither *in vitro* nor *in vivo* adjustments to free concentrations were applied. A full list of BER values derived across all method combinations

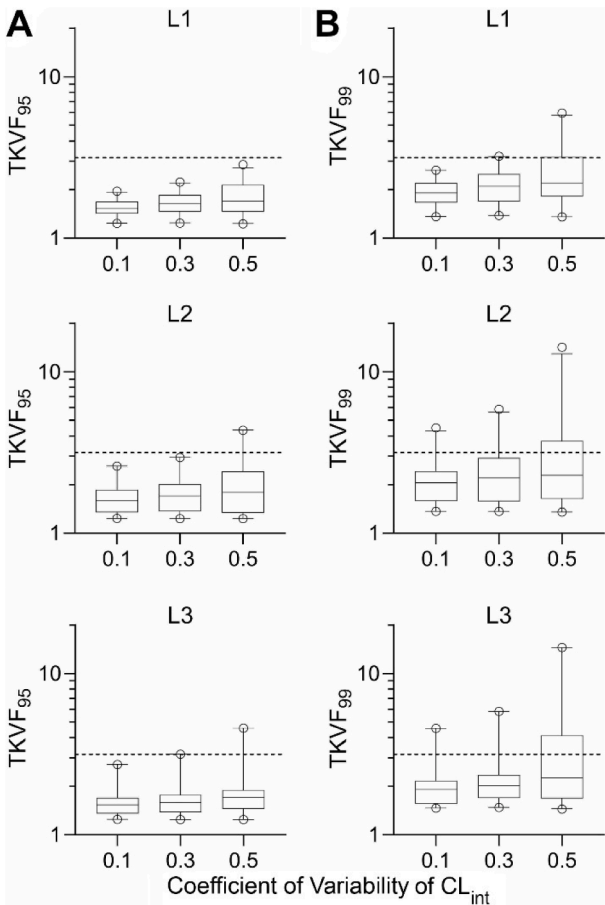
**Table 1**  
**Consistency between GastroPlus and *httk*  $C_{max}$  predictions.** Summary of the Spearman  $\rho$ , Pearson  $r$ , mean absolute error (MAE), and mean error (ME) values for comparing of  $C_{max}$  predicted using GastroPlus and the *httk* R package, for each of the three Levels of PBPK parameter integration applied (L1-L3 – see Fig. 1 and Methods). The 5 GastroPlus parameters referred to in (B) are logP, fraction unbound in plasma, blood-to-plasma ratio, hepatic intrinsic clearance, and the gut absorption rate ( $K_{gut\ abs}$ ).

| Level | Index  | <i>httk</i> model parameterization |                                     |                                             |                                                           |
|-------|--------|------------------------------------|-------------------------------------|---------------------------------------------|-----------------------------------------------------------|
|       |        | (A) Default <i>httk</i> parameters | (B) = (A) + 5 GastroPlus parameters | (C) = (B) without GastroPlus $K_{gut\ abs}$ | (D) = (B) + GastroPlus physiological parameters and $K_p$ |
|       |        | ●                                  | ◆                                   | ○                                           | □                                                         |
| L1    | MAE    | 0.48                               | 0.21                                | 0.28                                        | 0.39                                                      |
|       | ME     | 0.30                               | -0.03                               | 0.05                                        | 0.39                                                      |
|       | $r$    | 0.28                               | 0.78                                | 0.59                                        | 0.82                                                      |
|       | $\rho$ | 0.41                               | 0.68                                | 0.65                                        | 0.72                                                      |
| L2    | MAE    | 0.56                               | 0.27                                | 0.36                                        | 0.41                                                      |
|       | ME     | 0.47                               | 0.05                                | 0.15                                        | 0.37                                                      |
|       | $r$    | 0.60                               | 0.88                                | 0.77                                        | 0.85                                                      |
|       | $\rho$ | 0.56                               | 0.87                                | 0.75                                        | 0.82                                                      |
| L3    | MAE    | 0.63                               | 0.31                                | 0.38                                        | 0.47                                                      |
|       | ME     | 0.48                               | 0.21                                | 0.28                                        | 0.39                                                      |
|       | $r$    | 0.06                               | 0.88                                | 0.86                                        | 0.86                                                      |
|       | $\rho$ | 0.18                               | 0.80                                | 0.72                                        | 0.84                                                      |

is included in the Table S15.

The Point-Biserial correlation coefficient was calculated to evaluate the relationship between BER values and reported risk classifications. Fig. 5 and Fig. S3A show the Point-Biserial correlation coefficients between BER values and risk classifications, across different combinations of  $C_{max}$  sources, *in vitro* PODs, and whether  $f_{ub}$  adjustments were considered. Because differences in data availability of *in vitro* POD and  $f_{ub}$  across chemicals can influence the number of chemicals included in each BER calculation method, thereby affecting correlation performance, we also presented a restricted analysis with only the chemicals with PODs available from both SSBT and ToxCast datasets ( $n = 29$  for L1 and L2 PBK levels,  $n = 21$  for L3), shown in Fig. S3B. Based on the combined results from Fig. 5 and Fig. S3, we conclude that using the 95<sup>th</sup> percentile  $C_{max}$  from *httk-pop* slightly improves the performance in classifying risk levels, with marginally higher Point-Biserial correlation coefficients compared to point estimates from GastroPlus. In addition, *in vitro* PODs from SSBT resulted in the highest correlation coefficients. In Fig. S3B, when the number of chemicals was balanced, BERs based on ToxCast PODs showed similar performance to those using fixed PODs, suggesting that ToxCast-based PODs did not improve BER-informed risk classification. Finally, BERs based on nominal concentrations (without  $f_{ub}$  adjustment) have the highest Point-Biserial correlation values. When  $f_{ub}$  adjustments were applied, either using all predicted values or a combination of predicted and measured values, results were similar. This indicates that incorporating measured vs. modeled  $f_{ub}$  data did not improve BER-based classification performance for this set of chemicals and exposure scenarios.

In addition to correlation analysis, binary classification was also evaluated using a BER threshold of 1. As an illustration, for each exposure scenario and PBK parameter level, if the minimal BER exceeded 1, the chemical was classified as low risk; otherwise, as high risk. These BER-informed risk classifications were then compared to the reported classifications in the exposure assessment module, and binary classification metrics (accuracy, sensitivity, specificity, and MCC) were calculated. For example, red asterisks in Fig. 4 indicate cases where the minimal BER misclassified the risk level. As ToxCast-based PODs did not show better correlation performance than fixed PODs, binary classification metrics were only calculated for BERs based on fixed values and SSBT. To compare how different choices in BER calculation affect the

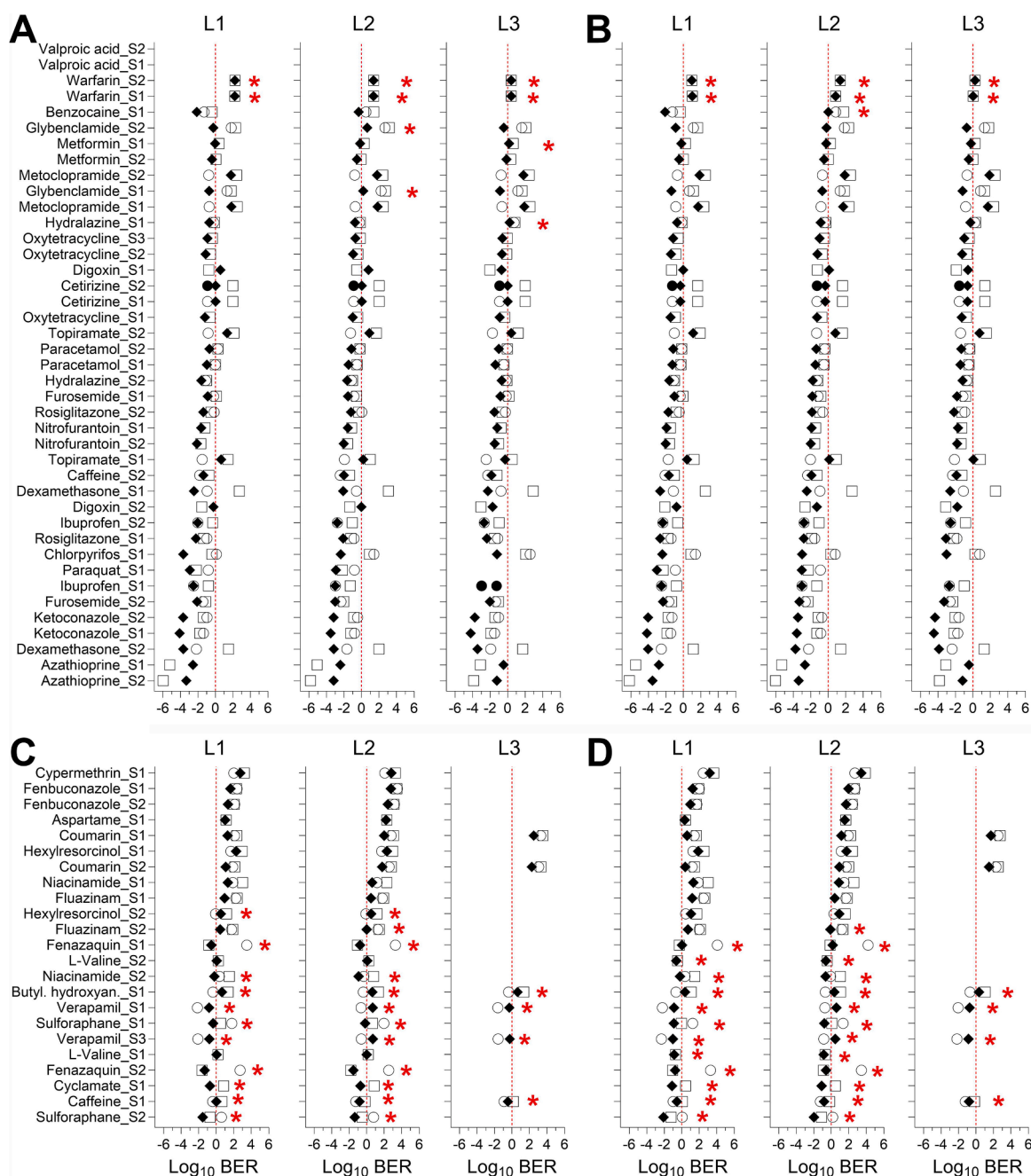


**Fig. 3. Inter-individual variability of  $C_{max}$  predicted using *httk-pop*.** (A)  $TKVF_{95}$  and (B)  $TKVF_{99}$  were calculated as the ratio of the median  $C_{max}$  to the 95<sup>th</sup> and 99<sup>th</sup> percentile  $C_{max}$ , respectively. Bars indicate the median; boxes represent the interquartile range (25<sup>th</sup> to 75<sup>th</sup> percentile); whiskers denote the 2.5<sup>th</sup> and 97.5<sup>th</sup> percentiles. Horizontal dashed lines indicate default TKVF of  $10^{1/2}$ .

accuracy, sensitivity, specificity, and MCC of risk classification, detailed values for these metrics were compiled in the Table S16. Among these metrics, MCC was emphasized because it has been recommended as a more comprehensive statistical measure that accounts for all four elements of the confusion matrix (i.e., true positives, false negatives, true negatives, and false positives) and is considered more informative than accuracy (Chicco & Jurman, 2020). Therefore, comparisons between different BER calculation methods were primarily based on MCC.

The trends observed in the binary classification comparisons were generally consistent with those from the correlation analysis. In terms of  $C_{max}$  simulation, BERs using *httk-pop* generally had slightly higher MCC values than those using GastroPlus. While GastroPlus with fixed PODs and L3 PBK parameterization sometimes performed better, it should be noted that the number of data points for low- and high-risk classifications at the L3 level is imbalanced. Using cell assay-derived PODs instead of fixed values generally improved MCC values, except in few specific cases (e.g., fixed POD = 0.01  $\mu$ M, L3 parameterization, and with  $f_{ub}$  adjustments). Similarly, not considering  $f_{ub}$  adjustments usually resulted in higher MCCs, except in a few cases with fixed PODs of 0.01  $\mu$ M.

Overall, to inform risk decisions based on BER using non-animal approaches, the approach with the most consistently high performance uses human-relevant cell assays to derive PODs without applying  $f_{ub}$  adjustments *in vivo* and *in vitro*. Conversely, the choice of kinetic simulation tools and whether population-level simulation is considered



**Fig. 4.** Bioactivity Exposure Ratio (BER) calculated using total *in vitro* PODs from the SSBT database (°: lowest  $AC_{50}$ ; □: cell stress; ◆: HTTr Bifrost) and predicted by either GastroPlus (A, C; total  $C_{\max}$  values) or *httk*-pop (B, D; the 95<sup>th</sup>-percentile  $C_{\max}$ ). Panels (A) and (B) show chemical and exposure scenario combinations classified as high risk, while panels (C) and (D) represent low-risk combinations. Red asterisks indicate cases where the minimal BER resulted in misclassification of the risk category.

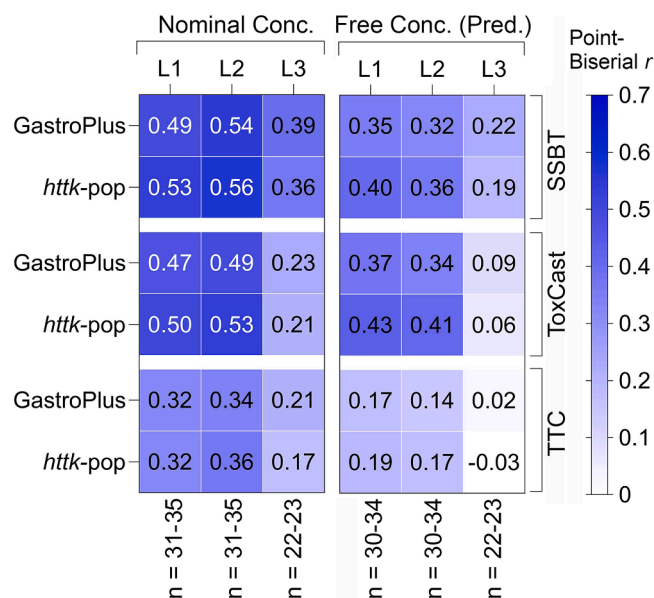
in the exposure assessment did not generally affect BER correlations with risk levels, although population-level simulation provided a slight improvement in some cases.

### 3.4. *In vitro-in vivo* concordance of OED

The *in vivo* human-equivalent PODs, extrapolated from animal PODs reported in Cable et al. (2025) and Aurisano et al. (2023), were used to evaluate the *in vitro-in vivo* concordance of OEDs. This evaluation was conducted both with and without  $f_{ub}$  adjustments. OEDs were converted using different *in vitro* POD sources, including the SSBT, ToxCast data, and fixed *in vitro* iTTC values (0.01 and 1  $\mu\text{M}$ ). Fig. 6 presents the *in vitro-in vivo* concordance results only using SSBT and ToxCast for a

dataset of 21 chemicals, which combines data from Cable et al. (2025) and Aurisano et al. (2023). More detailed concordance results, including additional POD sources, are shown in Fig. S5.

Based on the values of Spearman's  $\rho$ , Pearson's  $r$ , MAE, and ME shown in Fig. S5, we first observed that the chemical dataset (i.e., whether Aurisano et al. (2023) PODs were included) did not significantly impact the concordance. In addition, across all chemical sets, cell assay-derived PODs generally showed higher correlation coefficients ( $\rho$  and  $r$ ) compared to those using fixed iTTC values. However, the corresponding MAE and ME values were not the lowest when compared to those obtained using iTTCs of 0.01 and 1  $\mu\text{M}$ . Focusing specifically on the 21-chemical dataset and the use of cell assay-derived PODs (Fig. 6), concordance was generally improved when using PODs from SSBT



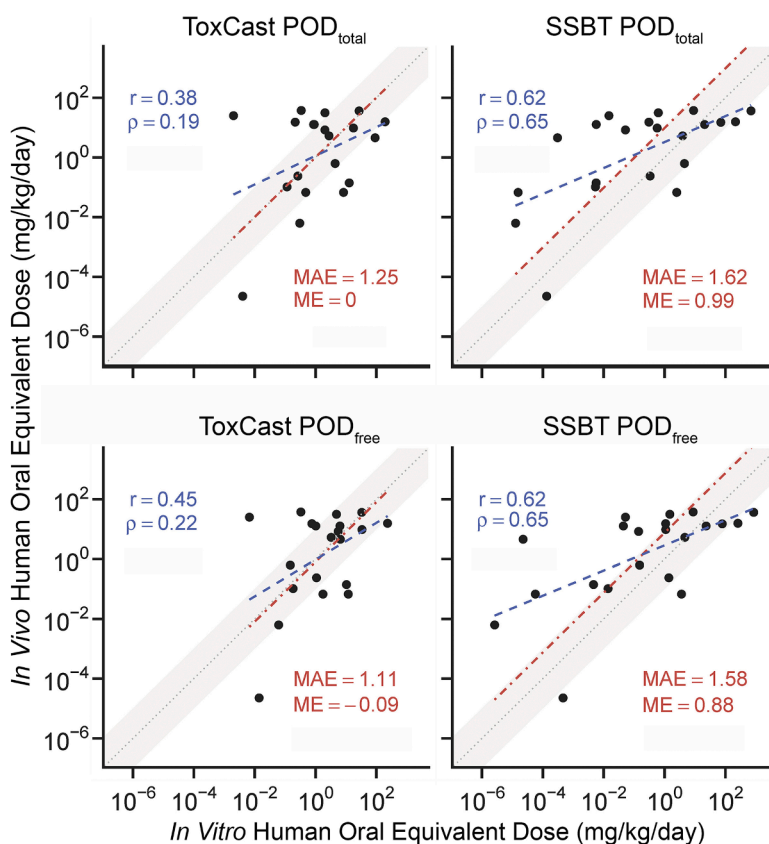
**Fig. 5.** Point-Biserial correlation coefficients (color and numbers inside each box) between (A) margin of exposure (MOE) or (B) bioactivity exposure ratio (BER), and the pre-assigned risk categories. See Methods for explanation of L1-L3 categories. The number of chemicals included in each correlation is indicated below each plot.

compared to ToxCast-based PODs. This was demonstrated by the higher correlation coefficients obtained when using SSBT. Additionally, applying  $f_{ub}$  adjustments only marginally improved concordance, slightly decreasing MAE and shifting ME values closer to zero.

## 4. Discussion

### 4.1. Strengthening scientific confidence in NGRA frameworks

With increasing regulatory interest in the use of NAMs to protect human health, building scientific confidence in the application of NGRA for regulatory use has become essential (European Chemicals Agency, 2024). One approach to demonstrate decision-making utility of NGRA is through case studies such as those presented in Middleton et al. (2022) and Cable et al. (2025). To evaluate the robustness of the NGRA framework used by these case studies, we conducted a sensitivity analysis focused on how variations in methodological choices across NGRA modules might influence safety conclusions. This sensitivity analysis aimed to identify method combinations that could enhance the performance and consistency of NGRA-based safety decision-making. Our findings support three primary outcomes of note for the regulatory use of NGRA. First, varying the choice of kinetic simulation tools or incorporating population-level variability in exposure modeling had modest impact on risk outcomes. Second, the use of human-relevant *in vitro* assays to derive points of departure (PODs), improved upon the performance of a toxicokinetics-only approach based on comparisons of predicted  $C_{max}$  with fixed iTTC values. Third, bioactivity exposure ratios (BERs) calculated without adjustment for *in vivo* or *in vitro*  $f_{ub}$  resulted in more conservative (i.e., health-protective) risk estimates without degradation in predictive performance. These outcomes support the use



**Fig. 6.** Scatter plots comparing oral equivalent doses (OEDs) derived from total and free *in vivo* human-equivalent and *in vitro* PODs using either ToxCast (left panel) or SSBT (right panel) datasets. “Total” concentrations refer to values without adjustment for the fraction unbound in plasma and media, while “free” concentrations account for both adjustments. Blue dashed lines represent the best-fit linear regression (used to calculate Pearson’s  $r$  and Spearman’s  $\rho$ ), and red dash-dot lines indicate the best-fit line with a fixed slope of 1 (used for calculating mean absolute error [MAE] and mean error [ME]).

of nominal dose metrics and human-based assays as the currently preferred approaches to strengthen BER-informed assessments.

#### 4.2. PBK modeling: reproducibility and efficiency

In the *exposure assessment* module, we compared  $C_{\max}$  simulated using the commercial software GastroPlus (as used in Middleton et al. (2022)) with those estimated by the open-source *httk* R package. Although the model structures differ in terms of the number of compartments that are simulated and the physiological parameters differ between the two platforms, we found that  $C_{\max}$  values from GastroPlus could be closely reproduced using *httk*, when matching key input parameters — namely  $\log P$ ,  $f_{ub,plasma}$ , blood-to-plasma ratio,  $CL_{int}$ , and gut absorption rate — to those that were also used in GastroPlus. These findings are consistent with previous reports (Prieto Garcia et al., 2022; Zhang et al., 2025), highlighting that platform choice does not substantially affect model output when appropriately parameterized. The ability of *httk* to generate  $C_{\max}$  estimates comparable to those from GastroPlus has important implications for high-throughput applications of oral exposure. The *httk* package, for example easily supports batch simulation of multiple chemicals and exposure scenarios (Pearce et al., 2017). Combined with its flexibility to adjust key parameters, *httk* offers a computationally efficient and scalable solution for population-based toxicokinetic modeling within NGRA workflows.

#### 4.3. Incorporating population variability to $C_{\max}$ estimates

Middleton et al. (2022) reasoned that considering interindividual variability in  $C_{\max}$  could be a potential improvement for NGRA to enhance safety decision-making. However, few *in vitro* models exist to experimentally derive such data (Rusyn et al., 2022); therefore, we used a modeling approach. Our comparison of  $C_{\max}$  predictions from GastroPlus (only simulated a specific individual of 30-year-old female with 60 kg body weight) and *httk-pop* (i.e., comparing to median  $C_{\max}$  with incorporating population variability) showed that simulations using *httk-pop* achieved slightly better agreement with GastroPlus, reflected by higher correlation coefficients and lower MAEs, demonstrating the advantage of modeling interindividual variability. However, incorporating the 95<sup>th</sup> percentile of  $C_{\max}$  into BER calculations did not lead to significant improvement in risk classification into low- or high-risk categories, with only marginal increases observed in Point-Biserial correlation coefficients. This limited effect may be due to the use of fixed CV values across all chemicals, which do not reflect true chemical-specific variability. As recommended by the World Health Organization/International Programme on Chemical Safety (WHO/IPCS, 2005), incorporating chemical-specific variability data represents a promising direction for future refinement. Population toxicokinetic variability can be quantified using data from virtual population-based TK models and *in vitro* cell experiments (Chiu, 2024; Lu et al., 2025; Zeise et al., 2013).

#### 4.4. Human-relevant *in vitro* assays improve BER performance

The selection of *in vitro* PODs strongly influenced the performance of BER-based risk assessments. A main advantage of NAMs lies in their capacity to generate mechanistically informative and human-relevant toxicity data, particularly when assays are based on human cells which reduces uncertainties associated with interspecies extrapolation and better capture human-specific biological responses (Schmeisser et al., 2023; Zink et al., 2020). Among the evaluated PODs, those derived from human-relevant assays (i.e., SSBT) produced the strongest correlations with historical regulatory classifications. These data outperformed both ToxCast-derived PODs and the fixed values based on the iTTC, in terms of both risk classification performance and *in vitro-in vivo* concordance for oral equivalent dose estimates. Although ToxCast offers a large number of assays and high throughput, it is still limited in

biological coverage and its inclusion of non-human and less tissue-specific assays may reduce human relevance. Moreover, the inclusion of gene expression data in SSBT may have enabled better coverage compared to other methods. Fixed PODs (e.g., 0.01 or 1  $\mu\text{M}$ ) offer simplicity — no new NAMs data are to be collected — but essentially result in an exposure-based classification that does not address toxicodynamics. These findings align with those of Lu et al. (2024), who demonstrated enhanced *in vitro-in vivo* concordance using human induced pluripotent stem cell (hiPSC)-derived models. Collectively, these findings highlight the robustness and potential utility of human-relevant *in vitro* assays for POD derivation, thereby strengthening BER calculations and supporting more informed and reliable safety decision-making.

#### 4.5. Impact of free vs. nominal *in vitro* dose metrics

The choice of a dose metric — whether to use nominal (total) or free concentrations *in vivo* and *in vitro* — has been discussed extensively in the context of NGRA and QIVIVE. Several publications reasoned that free concentrations in the *in vitro* assay media and *in vivo* plasma are more appropriate than nominal values, because they are more relevant to the molecular mechanisms underlying toxicity (Fisher et al., 2019; Heringa et al., 2004). However, our findings are contrary to this reasoning: adjusting for *in vitro* and *in vivo* free concentrations (i.e., using  $f_{ub}$  in both media and plasma) led to a reduction in the conservativeness of BER-based risk decisions. This observation aligns with findings by Henneberger et al. (2021) and Nicol et al. (2024), who suggested that using nominal concentrations often leads to more conservative predictions. Regarding the incorporation of  $f_{ub}$  into QIVIVE, our *in vitro*-to-*in vivo* concordance analysis for OED showed that using free concentrations resulted in only modest improvements. It is important to note that these findings are based on 35 chemicals occupying a relatively narrow physicochemical and pharmacokinetic space, and these parameters may influence how dose metrics perform in QIVIVE. Similarly, a study by Lin et al. (2025), which used PODs from a six-cell-type assay battery of 15 chemicals (also representing a limited chemical space) to estimate *in vivo* OEDs and compared them to reported regulatory values, found no substantial difference in concordance outcomes between nominal and free concentration approaches. These findings suggest that while free concentration metrics may offer mechanistic relevance, free concentration application to QIVIVE and BER calculations does not necessarily lead to more protective or predictive outcomes. Furthermore, as noted previously by Lin et al. (2025), there are a number of limitations with respect to existing mass balance models which may impact their ability to improve *in vitro*-to-*in vivo* concordance. For instance, both Lin et al. (2025) and present application of these models used “steady-state” predictions for *in vitro* concentrations, which may not accurately reflect *in vitro* kinetics across diverse assay systems and chemical properties. Recent work modeling chemical disposition in zebrafish embryo assays (which are part of ToxCast) suggest substantial time-dependence on the internal concentrations in the embryo (Rude et al., 2025). Thus, these results highlight the need for future research, including broader chemical datasets and incorporation of time-dependence, to fully characterize the potential impact of dose metric choice on QIVIVE.

#### 4.6. Study limitations

The sensitivity analyses detailed herein have several limitations. The original studies by Middleton et al. (2022) and Cable et al. (2025) included oral, inhalation, intravenous, and dermal exposure scenarios. However, due to limitations in the *httk* R package—specifically, the lack of a dermal PBK model and the limited flexibility to match inhalation and intravenous exposure settings used in GastroPlus—only the oral exposure route with relatively small set of chemicals (only 35) was evaluated in this study. Additionally, the *httk* R package does not

support modeling gut wall metabolism or transporter-mediated processes, which can be incorporated in GastroPlus through its ACAT model. This limitation may reduce the accuracy of predictions for compounds where intestinal metabolism or active transport plays a significant role (Korol-Bexell et al., 2025).

Furthermore, our comparison mainly based on  $C_{max}$  as the metric of internal exposure. However,  $C_{max}$  is sensitive to parameters such as the absorption rate and subject to variability from physiological factors including gastrointestinal transit time, gastric pH, and food intake (Kaus et al., 1999; Vinarov et al., 2021). Other pharmacokinetic metrics such as the area under the concentration-time curve (AUC) and average plasma concentration may offer informative and robust indicators of systemic exposure and clearance (Kaus et al., 1999; Smith et al., 2010). Future work could expand this framework to include AUC-based comparisons, thereby enhancing the comprehensiveness of cross-platform model evaluations.

Additionally, conducting this sensitivity analysis required a wide range of *in vivo/in vitro* toxicokinetic and toxicodynamic data (as shown in Fig. S2). For instance, for some assays we used default concentrations for FBS% in the media of *in vitro* assays when assay-specific information was not available, potentially reducing precision of our analyses comparing use of free vs. total concentrations. Different datasets also did not completely align in terms of chemicals, which affected the consistency of comparisons for risk classification performance. In particular, unequal numbers of data points across comparisons may influence the reliability of the Point-Biserial correlation coefficient. Furthermore, the limited availability of *in vivo* PODs from animal testing limited the number of data points usable for *in vitro*-to-*in vivo* concordance analyses.

Classification performance is dependent on the BER threshold selected. In this work we used a fixed BER threshold of 1 for illustration, however, future work could optimize the BER threshold to maximize, for example, protectiveness for each combination of inputs. While the approach taken here aligns with the study's objective of evaluating sensitivity to methodological choices, future work could benefit from incorporating combination-specific threshold analyses to assess protectiveness and utility. Despite these limitations, the sensitivity analyses still provide valuable insights into the robustness of NGRA frameworks and highlights areas for future methodological refinement.

#### 4.7. Conclusions and practical recommendations

This sensitivity analysis demonstrates the robustness and practical utility of the NGRA framework originally proposed by Baltazar et al. (2020). By systematically evaluating key methodological elements in the workflow, our findings highlight three main conclusions. First, different PBK simulation tools (GastroPlus vs. *httk*) produced comparable  $C_{max}$  estimates, supporting the feasibility of replacing commercial platforms with open source alternatives in high-throughput contexts. Second, PODs derived from human cell-based bioactivity data significantly improved BER-based risk classification, offering stronger consistency with historical regulatory benchmarks and enhancing the biological relevance of safety assessments. Third, using nominal concentrations, rather than adjusting for *in vivo* and *in vitro* free concentrations, produced more conservative and health-protective BERs without noticeable degradation in predictive performance.

Based on these findings, we propose the following recommendations as a practical starting point for NGRA implementation to other chemicals and decision contexts.

- Using customizable, open-source tools such as *httk*, with targeted adjustment of key chemical-specific parameters, can improve the transparency, reproducibility, and efficiency of PBK modeling.
- For BER calculations, we currently recommend the use of human-relevant *in vitro* NAMs and nominal concentration metrics because these approaches tend to result in more health-protective outcomes

and the incorporation of *in vitro* mass balance modeling has not yet been shown them to improve accuracy in risk characterization.

- Additionally, while incorporating modeling of population toxicokinetic variability may increase sensitivity, the impact may be limited in the absence of chemical-specific toxicokinetic variability data, particularly with respect to variability in metabolism.

Overall, this work highlights key methodological considerations for advancing the application of NGRA in regulatory contexts and promotes the use of NAM-based data in specific regulatory contexts (Lu et al., 2025). By promoting approaches that enhance human relevance, computational efficiency, and consistency, future implementations of NGRA can more effectively support public health protection goals while reducing reliance on animal testing.

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#### CRediT authorship contribution statement

**Hsing-Chieh Lin:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Maria T Baltazar:** Writing – review & editing, Resources, Project administration, Data curation, Conceptualization. **Sophie Cable:** Writing – review & editing, Investigation, Data curation, Conceptualization. **Lucie C. Ford:** Writing – review & editing, Investigation, Data curation. **Alistair M Middleton:** Writing – review & editing, Investigation, Data curation, Conceptualization. **Beate Nicol:** Writing – review & editing, Investigation, Data curation, Conceptualization. **Ans Punt:** Writing – review & editing, Investigation, Data curation, Conceptualization. **Joe Reynolds:** Writing – review & editing, Investigation, Data curation, Conceptualization. **Ivan Rusyn:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Weihsueh A Chiu:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

#### Declaration of competing interest

The following authors are employed by Unilever: Maria Teresa Baltazar, Alistair Middleton, Ans Punt, Sophie Cable, Beate Nicol, Joe Reynolds. Other authors declare no competing interests.

#### Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.namjnl.2025.100056](https://doi.org/10.1016/j.namjnl.2025.100056).

#### Data availability

Data is either included in Supplementary Materials or available on GitHub at [https://github.com/hsingchiehlin/Senana\\_NGRA](https://github.com/hsingchiehlin/Senana_NGRA) along with computer code.

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